

EXPERIMENT 2

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kali@kali: ~  
Session Actions Edit View Help  
Unknown -P option -Pn192.168.1.1.  
QUITTING!  
  
(kali@kali)-[~]  
$ nmap -pn192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 05:07 EDT  
Found no matches for the service mask 'n192.168.1.1' and your specified protocols  
QUITTING!  
  
(kali@kali)-[~]  
$ nmap -sn192.168.1.1  
Nmap 7.95 ( https://nmap.org )  
Usage: nmap [Scan Type(s)] [Options] {target specification}  
TARGET SPECIFICATION:  
  Can pass hostnames, IP addresses, networks, etc.  
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254  
  -iL <inputfilename>: Input from list of hosts/networks  
  -iR <num hosts>: Choose random targets  
  --exclude <host1[,host2][,host3], ...>: Exclude hosts/networks  
  --excludefile <exclude_file>: Exclude list from file  
HOST DISCOVERY:  
  -sL: List Scan - simply list targets to scan  
  -sn: Ping Scan - disable port scan  
  -Pn: Treat all hosts as online -- skip host discovery  
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports  
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes  
  -PO[protocol list]: IP Protocol Ping  
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]  
  --dns-servers <serv1[,serv2], ...>: Specify custom DNS servers  
  --system-dns: Use OS's DNS resolver  
  --traceroute: Trace hop path to each host  
SCAN TECHNIQUES:  
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans  
  -sU: UDP Scan  
  -sN/sF/sX: TCP Null, FIN, and Xmas scans  
  --scanflags <flags>: Customize TCP scan flags  
  -sI <zombie host[:probeport]>: Idle scan  
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans  
  -sO: IP protocol scan  
  -b <FTP relay host>: FTP bounce scan  
PORT SPECIFICATION AND SCAN ORDER:  
  -p <port ranges>: Only scan specified ports  
    Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
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  Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9  
  --exclude-ports <port ranges>: Exclude the specified ports from scanning  
  -F: Fast mode - Scan fewer ports than the default scan  
  -r: Scan ports sequentially - don't randomize  
  --top-ports <number>: Scan <number> most common ports  
  --port-ratio <ratio>: Scan ports more common than <ratio>  
SERVICE/VERSION DETECTION:  
  -sV: Probe open ports to determine service/version info  
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)  
  --version-light: Limit to most likely probes (intensity 2)  
  --version-all: Try every single probe (intensity 9)  
  --version-trace: Show detailed version scan activity (for debugging)  
SCRIPT SCAN:  
  -sC: equivalent to --script=default  
  --script=<Lua scripts>: <Lua scripts> is a comma separated list of  
    directories, script-files or script-categories  
  --script-args=<n1=v1,[n2=v2,...]>: provide arguments to scripts  
  --script-args-file=filename: provide NSE script args in a file  
  --script-trace: Show all data sent and received  
  --script-updatedb: Update the script database.  
  --script-help=<Lua scripts>: Show help about scripts.  
    <Lua scripts> is a comma-separated list of script-files or  
    script-categories.  
OS DETECTION:  
  -O: Enable OS detection  
  --osscan-limit: Limit OS detection to promising targets  
  --osscan-guess: Guess OS more aggressively  
TIMING AND PERFORMANCE:  
  Options which take <time> are in seconds, or append 'ms' (milliseconds),  
  's' (seconds), 'm' (minutes), or 'h' (hours) to the value (e.g. 30m).  
  -T<0-5>: Set timing template (higher is faster)  
  --min-hostgroup/max-hostgroup <size>: Parallel host scan group sizes  
  --min-parallelism/max-parallelism <numprobes>: Probe parallelization  
  --min-rtt-timeout/max-rtt-timeout/initial-rtt-timeout <time>: Specifies  
    probe round trip time.  
  --max-retries <tries>: Caps number of port scan probe retransmissions.  
  --host-timeout <time>: Give up on target after this long  
  --scan-delay/--max-scan-delay <time>: Adjust delay between probes  
  --min-rate <number>: Send packets no slower than <number> per second  
  --max-rate <number>: Send packets no faster than <number> per second  
FIREWALL/IDS EVASION AND SPOOFING:  
  -f; --mtu <val>: fragment packets (optionally w/given MTU)  
  -D <decoy1,decoy2[,ME], ...>: Cloak a scan with decoys
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Session Actions Edit View Help  
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output  
MISC:  
-6: Enable IPv6 scanning  
-A: Enable OS detection, version detection, script scanning, and traceroute  
--datadir <dirname>: Specify custom Nmap data file location  
--send-eth/--send-ip: Send using raw ethernet frames or IP packets  
--privileged: Assume that the user is fully privileged  
--unprivileged: Assume the user lacks raw socket privileges  
-V: Print version number  
-h: Print this help summary page.  
EXAMPLES:  
nmap -v -A scanme.nmap.org  
nmap -v -sn 192.168.0.0/16 10.0.0.0/8  
nmap -v -iR 10000 -Pn -p 80  
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES  
Scantype 1 not supported  
  
(kali@kali)-[~]  
$ nmap -PR192.168.1.1  
Unknown -P option -PR192.168.1.1.  
QUITTING!  
  
(kali@kali)-[~]  
$ nmap -pR192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 05:09 EDT  
Error #485: Your port specifications are illegal. Example of proper form: "-100,200-1024,T:3000-4000,U:60000-"  
QUITTING!  
  
(kali@kali)-[~]  
$ nmap -n 192.168.1.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 05:10 EDT  
Nmap scan report for 192.168.1.1  
Host is up (0.020s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 4.49 seconds
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